Nirbhay Singh

AWS DevOps Architect | AWS Solutions Architect - Professional

11+	$45K+	30x	11,000+	14+
Years Experience	AWS Cost Savings	CI/CD Speedup	Servers Migrated	Certifications

PROFESSIONAL SUMMARY

AWS-certified DevOps Architect (Solutions Architect Professional) with 11+ years designing, automating, and operating production-grade AWS infrastructure at enterprise scale. Deep expertise across the full AWS stack: EKS, EC2, RDS, S3, VPC, Batch, Spot Instances, SageMaker, Lambda, and CloudFormation. Proven record of reducing AWS costs by $45K+, accelerating CI/CD by 30x, orchestrating 1,000+ containers on EKS, and migrating 11,000+ servers to AWS. Currently leading AWS architecture strategy at Bosch Europe with EUR 4M+ in documented cloud savings.

KEY ACHIEVEMENTS

- AWS Solutions Architect Professional certified with 8+ years of continuous AWS production experience across 4 enterprises
- Reduced AWS costs by $45,000 at Deloitte through Spot Instance strategies, Reserved Instances, right-sizing, and pipeline optimisation
- Orchestrated 1,000+ containers using AWS Batch, Spot Instances, and EKS for distributed analytics and ML workloads
- Migrated 11,000+ server on-premise infrastructure to AWS with zero-downtime cutover and automated rollback strategies
- Designed fault-tolerant AWS architectures (multi-AZ, cross-region) with 99.99% uptime SLAs across production environments
- Built data & AI platforms on SageMaker, EMR, and Redshift serving Fortune 500 analytics workloads
- Created InfraCents (open-source): real-time AWS cost estimation on every Terraform pull request

PROFESSIONAL EXPERIENCE

Cloud & AI Architect (AWS Lead)

Dec 2022 - Present

Robert Bosch (Bosch Polska) Warsaw, Poland

- Led strategic AWS partnership for Bosch Poland -- defined AWS architecture standards, landing zone design, and multi-account governance using AWS Organizations and SCPs
- Designed AWS-native AI/ML infrastructure: SageMaker pipelines, EKS model serving, S3 data lake, and Glue ETL for enterprise workloads
- Implemented FinOps governance for 50+ AWS accounts, contributing to EUR 4M+ total savings through Savings Plans, right-sizing, and automated cleanup
- Established AWS Well-Architected reviews as standard practice -- mentored architects on security, reliability, and cost optimisation pillars

Senior DevOps Consultant

Jun 2021 - Nov 2022

McAfee

- Automated AWS infrastructure with Terraform (300+ resources), achieving 50%+ reduction in deployment time with near-zero configuration drift
- Deployed and operated production EKS and ECS clusters with auto-scaling, spot integration, and GitOps-driven deployments via ArgoCD
- Built AWS observability: CloudWatch dashboards, custom metrics, Prometheus, and Grafana for real-time infrastructure and application monitoring
- Implemented AWS security best practices: IAM least-privilege, KMS encryption, VPC flow logs, and automated compliance scanning

Senior DevOps Engineer / Data & AI Architect

Dec 2018 - Jun 2021

Deloitte

- Reduced AWS costs by $45,000 through Spot Instance fleet management, Reserved Instances, Savings Plans, and workload right-sizing

Nirbhay Singh

AWS | EKS | Terraform | CI/CD
CloudFormation | SageMaker | Cost Optimisation

CONTACT

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CORE SKILLS

AWS COMPUTE

EC2, EKS, ECS, Lambda, Batch, Spot Instances, ASG, Fargate

AWS STORAGE & DB

S3, RDS, Redshift, DynamoDB, EBS, EFS, Glacier, ElastiCache

AWS NETWORKING

VPC, Route53, CloudFront, ALB/NLB, Transit GW, Direct Connect, PrivateLink

AWS AI/ML & DATA

SageMaker, EMR, Glue, Athena, Kinesis, Redshift, Lake Formation

AWS SECURITY

IAM, KMS, CloudTrail, GuardDuty, Security Hub, Config, Organizations, SCPs

IAC & CI/CD

Terraform, CloudFormation, CDK, GitHub Actions, Jenkins, CodePipeline, ArgoCD

CONTAINERS

EKS, ECR, ECS Fargate, Docker, Helm

OBSERVABILITY

CloudWatch, X-Ray, Prometheus, Grafana, ELK

LANGUAGES

Python, Bash, YAML, JSON, HCL, PowerShell

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

- Accelerated CI/CD pipeline speed by 30x through CodePipeline architecture redesign, parallel stages, and artefact caching
- Orchestrated 1,000+ containers using AWS Batch, Spot Instances, and EKS for distributed data processing and ML training
- Built data & AI platforms on EMR, Redshift, and SageMaker -- serving analytics and model training for Fortune 500 clients
- Implemented AWS security controls: IAM policies, KMS encryption, CloudTrail auditing, GuardDuty threat detection, and Config rules

Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

PTC

- Designed secure, fault-tolerant AWS architectures (EC2, RDS Multi-AZ, S3, VPC) for production SaaS workloads
- Led migration of 11,000+ server on-premise infrastructure to AWS -- zero-downtime cutover with automated rollback
- Built disaster recovery with cross-region replication, automated failover, and RTO/RPO validation testing
- Managed AWS networking: VPC design, subnets, security groups, NACLs, Transit Gateway for multi-tier applications

System Engineer

Mar 2013 - Jul 2014

IBM

- Infrastructure operations, P1/P2 incident response, and root cause analysis for enterprise production environments

OPEN-SOURCE PROJECTS

InfraCents -- AWS Cost Estimates on Every Terraform PR

>> github.com/nirbhays/infracents

GitHub App posting real-time AWS cost estimates on Terraform pull requests. Parses TF changes, queries live AWS Pricing API. Zero config, native PR integration.

STACK: Python, Next.js, Terraform, GitHub App, AWS Pricing API

ShieldIaC -- AI-Powered IaC Security Scanner

>> github.com/nirbhays/shieldiac

Scans Terraform/CloudFormation for AWS security misconfigs pre-production. 100+ rules, 9 compliance frameworks (CIS AWS Benchmark, SOC2, HIPAA, PCI-DSS), GPT-4.1 fix suggestions.

STACK: Python, Terraform, CloudFormation, GPT-4.1, GitHub Actions

TokenMeter -- LLM Cost Intelligence

>> github.com/nirbhays/tokenmeter

Tracks token cost and latency across LLM providers. Smart routing saves up to 60% on AI spend with <5ms overhead.

STACK: Python, OpenAI, Anthropic, Google AI, Smart Routing, PyPI

CERTIFICATIONS & CREDENTIALS

- **AWS** Solutions Architect - Professional (Feb 2020)
- **CKAD** - Certified Kubernetes Application Developer (Valid Sep 2026)
- **Google** Professional Cloud Architect (Valid Jun 2026)
- **Google** Professional ML Engineer (Valid Dec 2027)
- **Google** Associate Cloud Engineer (Aug 2023)
- **AWS** Solutions Architect - Associate (Jan 2021)
- **HashiCorp** Terraform Associate
- **Google** Professional Cloud DevOps Engineer (Valid Dec 2027)
- **Google** Generative AI Leader (Valid Aug 2028)
- **MCSA** - Microsoft Certified Solutions Associate (Apr 2015)